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Complex Carbonate Reservoirs, Field Cases of Improving Recovery via Innovative Chemical Deep Conformance and Injectivity Management Technologies

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Abstract

Significant potential for higher recovery remain in complex carbonate reservoirs especially where water flood sweep efficiency is challenging. This paper share field cases of chemical (especially around polymer gel) deep conformance control, water injection well injectivity improvements (non-mechanical). The results of a large number of recent applications from carbonate reservoirs operated by CNPC affiliates or CNPC partners are reviewed. Challenges and risks are highlighted. Recommendations for unlocking complex carbonate reservoir potential via improving water flood sweep and injectivity using chemical approach are made.

Introduction

Water flooding, especially in carbonate reservoirs, faces challenges of managing water production. In mature oil fields, water cut can exceed 50%. This issue is particularly acute in China, where many of the country's largest oil fields, such as field D*, S*, and C*, have entered advanced stages of development characterized by declining oil production and sustained high water production. The economic and operational complexity associated with high water production are substantial, including injection water supply constraints, increased lifting costs, corrosion and scaling in production equipment, and in some cases environmental challenges of handling and disposing of large volumes of produced water. Where reservoir geology is complex, fast pathways between injection and production wells may exist, leading to short circuit of injection water from injector to producers. Potential consequences may include reduced water injection efficiency, and poor reservoir sweep. In response to these challenges, chemical deep conformance control technologies have emerged as a critical strategy for improving reservoir sweep efficiency, reducing water production, and enhancing ultimate oil recovery.

Chemical deep conformance control differs fundamentally from conventional near-wellbore water shutoff treatments. While traditional methods focus on sealing water influx near the production well using cement, mechanical isolation, or polymer gels, chemical deep conformance control targets the reservoir itself, modifying fluid flow paths deep within the formation. This approach involves the injection of

specialized chemical systems - such as specialty gels, microspheres, foams, and pH- sensitive polymers - that selectively block high-permeability zones or fractures, diverting injected water into previously unswept regions of the reservoir. By addressing water channeling or fast pathways at its source, chemical deep conformance control not only reduces water production but also improves the overall efficiency of waterflooding operations, delays premature well abandonment, and maximizes the recovery of remaining oil reserves.

In some of the large oil fields in China, where waterflooding has been the dominant secondary recovery method for decades, the need for effective deep conformance control is especially pressing. These major oil fields exhibit a wide range of geological complexities, including high heterogeneity, low and high permeability with large contrast, and extensive fault systems, all of which can contribute to uneven waterflood sweep and early water breakthrough. For instance, the D* oilfield, one of the most prolific, has seen water cuts rise above 90% in many blocks, necessitating advanced deep conformance control measures to sustain production. Similarly, the S* oilfield, with its complex faulted reservoirs, and the C* oilfield, characterized by presence of ultra-low permeability, present unique challenges that require tailored chemical deep conformance solutions.

The development and application of chemical deep conformance control technologies in China have been driven by both necessity and innovation. Early efforts in the 1980s and 1990s focused on polymer flooding and near-wellbore gel treatments, but these methods often provided only relatively temporary relief. Over time, advancements in polymer chemistry, nanotechnology, and reservoir engineering have led to more sophisticated solutions capable of penetrating deeper into the formation and providing longer-lasting conformance improvement. Today, CNPC operated or partner-operated oilfields is at the forefront of research and field application of chemical deep conformance control technologies, with extensive pilot and full-field implementations demonstrating significant improvements in oil recovery and water production management.

This paper provides an examination of chemical deep conformance control technologies as applied in CNPC operated or partner-operated oil fields. It begins with a detailed exploration of the challenges posed by high water production in China's major oil fields, highlighting the geological and operational factors that make conformance control essential. The paper then traces the historical development of conformance control technologies, from early polymer flooding to modern smart gels and nanofluid systems. Subsequent sections delve into the key technologies currently in use, including polymer gels, microspheres, foams, and pH-sensitive systems, with detailed case studies from D*, S*, C*, and other fields. The paper also discusses emerging trends and future directions, such as intelligent gel systems, nanotechnology enhancements, and data- driven optimization, which promise to further improve the effectiveness of deep conformance control. Finally, the paper concludes with an assessment of the economic and environmental implications of these technologies.

Technology Development in China, an Overview

The evolution of chemical deep conformance control technologies represents a significant advancement in water production management and oil recovery maximisation. A short historical analysis traces the technological progression from early polymer flooding experiments to today's sophisticated smart gel systems, with particular attention to the chemical innovations, field applications, and lessons learned are presented. The broad spectrum of technologies contributed towards best in class reservoir management practices.

Early Foundations (early years --1970s): The Polymer Flooding Era

The origins of chemical conformance control can be traced to the 1960s when petroleum engineers first recognized the potential of water-soluble polymers to improve oil recovery. During this period, the primary

focus was on polymer flooding - the injection of viscous polymer solutions to improve the mobility ratio between injected water and reservoir oil. The first generation of polymers used in China's D* Oilfield and other major basins consisted of partially hydrolyzed polyacrylamide (HPAM) with molecular weights ranging from 5-15 million Daltons.

These early polymer floods demonstrated two critical phenomena that would later form the basis of deep conformance control: viscosity enhancement and permeability reduction. Field tests showed that while the primary mechanism was viscosity modification to improve sweep efficiency, the polymers also exhibited adsorption and retention in high- permeability zones, leading to a disproportionate permeability reduction (RRF) effect. This incidental observation of permeability modification sparked interest in developing dedicated conformance control systems.

The chemical formulations of this era were relatively simple. A typical 1970s polymer flood in D* field S reservoir used HPAM at concentrations of 800-1200 mg/L in injection water, with no crosslinking agents. Field results were promising but inconsistent - some patterns showed 8-10% incremental recovery, while others showed minimal improvement due to mechanical degradation of polymers in high-permeability channels.

The Crosslinking Revolution (1980s-1990s)

The 1980s marked a paradigm shift with the introduction of crosslinked polymer systems specifically designed for conformance control. Researchers discovered that by adding metal ions or organic crosslinkers to polymer solutions, they could create three-dimensional gel networks with tunable strength and gelation times.

Chromium (Cr^{3+}) crosslinking emerged as the dominant technology during this period. The most common formulation combined HPAM (0.3-0.8% concentration) with chromium acetate or chromium propionate (100-500 mg/L) at reservoir temperatures of 45-90°C. The gelation chemistry involved ligand exchange reactions where Cr^{3+} ions coordinated with carboxylate groups ($-\text{COO}^-$) on the HPAM chains, forming stable octahedral complexes.

Field applications revealed several critical insights about gel behavior:

- Gelation time could be controlled from hours to weeks by adjusting pH (typically 4.5-6.5) and crosslinker concentration
- Gel strength depended on polymer molecular weight and degree of hydrolysis
- In-depth placement required careful design of injection slugs to prevent premature gelation.

The late 1980s saw the development of organic crosslinkers as alternatives to chromium systems. Polyethyleneimine (PEI)- based gels offered several advantages, including reduced toxicity and better performance in high-salinity environments. A typical PEI gel formulation for Daqing's high-temperature reservoirs (75°C) used 0.5% HPAM with 0.1% PEI crosslinker, achieving gelation times of 3-7 days.

The Diversification Period (1990s-2000s)

The 1990s witnessed an explosion of new conformance control technologies as researchers addressed the limitations of conventional gels. Three major advancements characterized this period:

Microsphere Technology Development. The concept of injectable microspheres emerged as a solution to the depth-of-penetration limitations of bulk gels. Early microspheres were based on crosslinked polyacrylamide with particle sizes ranging from 0.1-10 microns. These particles would swell 5-20 times their original size upon contact with formation water, selectively plugging high-permeability zones. S* oilfield's 1998 pilot test demonstrated that 2-micron microspheres could penetrate 30-50 meters into the formation before swelling and creating flow barriers.

The success of a microgel system depends on precise chemical engineering. Particle Size: must be smaller than the pore throat diameters of the target rock to ensure deep injection without immediate filtration. Swelling Ratio: the degree to which the microgel expands upon activation. This must be optimized to create an effective plug without causing near-wellbore screen-out. Kinetics: the rate of swelling must be carefully controlled - slow enough to allow deep placement but fast enough to be practical for the field project timeline. Trigger Mechanism: the design of the salinity-, temperature-, or time-dependent trigger is the core of the "intelligence" of the system. It must be tailored to the specific brine chemistry and temperature of the target reservoir. Mechanical & Chemical Stability: the swollen microgels must be strong enough to withstand reservoir pressure and not degrade over time.

Deep Particle Gels (DPGs). DPGs are pre-formed, cross-linked polymer particles with a specific size and mechanical strength designed for deep reservoir placement.

- Composition: They are typically made from cross-linked polyacrylamide or acrylamide copolymer gels.
- Form: They are not injected as a liquid solution. They are pre-formed at the surface, meaning the chemical crosslinking reaction that creates the gel particle is completed before injection.
- Appearance: They resemble small, soft, water-swellaable beads or granules (often called "gel pills"). They are supplied dehydrated or in a pre-swollen state.

The success of a DPG treatment hinges on designing the particles correctly for the specific reservoir. For example, Particle Size Distribution (PSD), this is the most critical factor. The PSD must be meticulously matched to the pore throat size distribution of the thief zone. Particles must be large enough to plug the target, but small enough to be injected and propagate without causing immediate screen-out at the wellbore. Swelling Ratio: the ability of the dehydrated particles to absorb water and swell to a specific size. This allows for easier injection (pump smaller particles) that then expand downhole to the required plugging size. Mechanical Strength (Elasticity/Strength): the gel particles must be strong enough to withstand the high shear forces during injection and the high reservoir closure stresses without breaking apart. However, they must also be deformable enough to navigate through the pore network. Thermal/Chemical Stability: they must remain stable (not degrade, dissolve, or shrink) at the reservoir's temperature and in its brine chemistry for a long enough period to be economically successful (often months to years)

Foam Conformance Control. Foam systems gained prominence for their ability to combine conformance control with gas flooding. The technology relied on surfactant-stabilized gas bubbles (typically nitrogen or CO₂) that preferentially entered water-swept zones. Key developments included:

- Fluorocarbon surfactants for high-temperature stability
- Foam quality optimization (60-85% gas fraction)
- Foam durability enhancers like nanoparticles
- Environmentally Adaptive Gels.

The late 1990s saw the first generation of "smart" gels that could respond to reservoir conditions. pH-sensitive gels using copolymers of acrylamide and acrylic acid demonstrated reversible swelling behavior. Temperature-activated systems based on poly(N-isopropylacrylamide) showed promise for deep reservoirs.

The Nanotechnology Era (2010s-Present)

The current generation of conformance control technologies leverages nanotechnology and advanced materials science to overcome historical limitations:

Nanogel Systems. Silica-based nanogels with particle sizes of 10-100 nm can penetrate ultra-low permeability zones (<10 mD) that were previously inaccessible. Surface functionalization with amine or carboxyl groups allows controlled aggregation in target zones.

Intelligent Gel Systems. Modern responsive gels incorporate multiple triggers:

- Redox-sensitive gels using ferric ion crosslinks
- Enzyme-activated systems for delayed gelation
- Magnetic nanoparticle-containing gels for external control

Hybrid Technologies. The most advanced systems combine multiple mechanisms:

- Microsphere-gel composites for staged conformance
- Foam-assisted nanogel delivery systems
- CO₂-responsive polymers for combined conformance control and miscible flooding

Field results from recent applications demonstrate the remarkable progress made. D* oilfield 2021 pilot using temperature- sensitive nanogels achieved a 28% reduction in water cut compared to 12% with conventional gels. The system used poly (NIPAM-co-AMPS) nanogels with a volume phase transition temperature of 65°C, precisely matching the reservoir conditions.

Technological Challenges and Solutions in Chemical Deep Conformance

The development and implementation of chemical deep conformance control technologies have faced numerous technical challenges throughout their evolution. These obstacles have driven significant innovation in polymer chemistry, fluid mechanics, and reservoir engineering. The industry's response to these challenges has shaped current conformance control practices and continues to influence research directions today.

Depth of Penetration and Placement Control

One of the most persistent challenges in deep conformance control has been achieving sufficient penetration of chemical treatments into the reservoir. Early gel systems often formed plugs near the wellbore due to rapid gelation, failing to address water channeling deep in the formation. This limitation severely restricted treatment effectiveness, particularly in thick reservoirs or those with extensive fracture networks.

The solution to this challenge emerged through multiple technological pathways. Delayed crosslinking systems represented a major breakthrough, with chromium malonate and chromium propionate formulations offering gelation times adjustable from several days to weeks. These systems allowed the treatment fluid to penetrate tens of meters into the formation before developing full gel strength. Field applications demonstrated that gelation time could be precisely tuned by controlling pH and crosslinker concentration - for instance, in D* oilfield's PIX-X layer, maintaining pH at 5.5-6.0 extended gelation time to 7-10 days, enabling treatment of zones 30-80 meters from injection wells.

In-situ crosslinking systems provided another solution, where the crosslinker formed gradually through chemical reactions. The aluminum citrate system, first applied in S* Oilfield in the late 1990s, generated crosslinking ions slowly through citrate decomposition at reservoir temperatures (65-85°C). This approach achieved penetration depths exceeding 50 meters while maintaining controllable gelation kinetics.

Microsphere technology offered a fundamentally different solution to the penetration challenge. By using pre-formed particles that could migrate through porous media before swelling, engineers could achieve deep placement without relying on delayed chemical reactions. The development of size-controlled microspheres

(1-100 gm) with tunable swelling ratios (5-20x) allowed customized treatments for different reservoir conditions. C* oilfield's ultra-low permeability reservoirs benefited particularly from 5-10 gm microspheres that could penetrate tight formations before expanding to create effective flow barriers.

Advanced placement techniques further improved treatment effectiveness. Tracer-guided injection profiling, using chemical tracers, enabled real-time monitoring of fluid distribution. Diversion agents, including temporary gels or foams, helped ensure treatment fluids reached target zones in heterogeneous reservoirs. Real-time pressure monitoring during injection provided immediate feedback on fluid placement and formation response.

Gel Stability and Longevity

Maintaining conformance control effectiveness over extended periods presented another major challenge. Early gel systems often suffered from syneresis (water expulsion), microbial degradation, or thermal instability, leading to premature treatment failure.

Syneresis proved particularly problematic in high-temperature reservoirs, where gels could lose up to 60% of their original volume within months. Researchers addressed this through polymer backbone modifications, introducing monomers like 2- acrylamido-2-methylpropane sulfonic acid (AMPS) that improved water retention. Field tests in China Xinjiang's high- temperature (90°C) reservoirs showed AMPS-modified gels maintained over 80% of their original volume after 12 months, compared to 40% for conventional HPAM gels.

Microbial degradation emerged as a significant issue in reservoirs with active microbial populations. The D* oilfield experience demonstrated that untreated gels could lose 50-70% of their viscosity within 6 months due to bacterial action. The solution came through several approaches: incorporation of biocides like glutaraldehyde or isothiazolinones, development of bioresistant polymer backbones (e.g., incorporating acrylamide tertiary butyl sulfonic acid (ATBS)), and oxygen scavengers to create anaerobic conditions unfavorable to microbial growth.

Thermal stability represented another critical challenge, especially for deep, hot reservoirs. Conventional polyacrylamide- based systems can mostly began degrading at temperatures above 60°C. The development of thermally stable polymers incorporating N-vinylpyrrolidone (NVP) and AMPS monomers extended the operational range to 120°C. China Tarim Basin applications demonstrated these advanced polymers maintained effective conformance control for a number of years in 110°C reservoirs.

Heterogeneity Management

Reservoir heterogeneity posed perhaps the most complex challenge for chemical deep conformance control. Extreme permeability contrasts, fracture networks, and layered formations could cause treatment fluids to bypass target zones entirely.

The industry responded with several innovative solutions. Adaptive injection strategies using real-time monitoring and control allowed adjustment of treatment parameters during execution. In L* Oilfield's heavy oil reservoirs, engineers employed step-rate injection tests during treatment to identify and preferentially treat high-permeability zones. This approach improved fluid distribution compared to conventional bullhead injection.

Advanced diversion techniques provided another solution. Viscoelastic surfactant-based diversion fluids could temporarily block treated zones, forcing subsequent treatment stages into new intervals. The S* experience showed that combining temporary diversion with microsphere pre-flushes improved zonal coverage from 40% to over 70% in highly stratified reservoirs.

Nanotechnology offered new possibilities for treating ultra-heterogeneous formations. Silica nanoparticles functionalized with responsive polymers could migrate through varying permeability zones, then aggregate selectively in target areas when triggered by temperature, pH, or ionic strength changes. Pilot

tests in J* oilfield demonstrated that these "smart" nanoparticles could improve conformance in formations with permeability contrasts exceeding 100:1.

Environmental and Safety Considerations

As environmental regulations tightened, conventional conformance control chemicals faced increasing scrutiny. Chromium- based systems, while effective, raised concerns about toxicity and potential groundwater contamination.

The industry developed several alternatives. Organic crosslinkers based on polyethyleneimine (PEI) or phenol- formaldehyde provided comparable performance without heavy metals. Bio-based gels using modified polysaccharides (e.g., scleroglucan or xanthan gum derivatives) offered biodegradable options for environmentally sensitive areas. D* oilfield recent trials with chitosan-based gels demonstrated effective conformance control with minimal environmental impact.

Safety concerns also drove innovation in handling and application. The development of pre-mixed, stabilized gel systems reduced onsite chemical handling risks. Encapsulated crosslinkers that activated only at reservoir temperatures improved worker safety during surface operations.

Economic Optimization

Cost-effectiveness remained a constant challenge, particularly in low oil price environments. The solution involved both technological improvements and operational innovations.

Chemical optimization reduced required treatment volumes without sacrificing effectiveness. The development of high- performance polymers (e.g., hydrophobically associating polyacrylamides) allowed lower concentrations (0.1-0.3% vs. 0.50.8% for conventional systems) to achieve similar results. Microsphere technology provided particularly favorable economics, with treatment costs 30-50% lower than conventional gels in comparable applications.

Operational improvements included:

- Just-in-time mixing systems that reduced chemical waste
- Selective zone treatment using advanced placement techniques
- Treatment design optimization through reservoir simulation

The integration of these approaches has made deep conformance control economically viable even in marginal fields. Recent applications in mature waterfloods have demonstrated payback periods under 18 months, with incremental recovery often exceeding 10%.

Monitoring and Evaluation

Assessing treatment effectiveness presented significant technical challenges. Conventional production logging often couldn't distinguish between conformance control effects and natural reservoir changes.

The industry developed several solutions:

- Tracer technology using unique chemical or isotopic signatures
- Time-lapse pressure transient analysis
- Distributed temperature sensing (DTS) and acoustic monitoring
- Advanced reservoir simulation incorporating treatment effects

These monitoring techniques not only improved post-treatment evaluation but also enabled real-time adjustment of ongoing treatments. The integration of monitoring data with machine learning algorithms has begun providing predictive capabilities for treatment optimization.

Future Challenges and Emerging Solutions

While significant progress has been made, new challenges continue to emerge. Ultra-deep reservoirs (below 4,000m) present higher temperature and pressure conditions that test material limits. Unconventional reservoirs require novel approaches to address complex fracture networks. Environmental regulations continue to tighten, demanding greener chemistries.

Emerging solutions include:

- High-temperature nanocomposite gels stable to 150°C
- Biomimetic polymers that self-repair mechanical damage
- CO₂-responsive systems for combined conformance control and carbon storage
- Machine learning-driven treatment design and optimization

The ongoing evolution of conformance control technology demonstrates how fundamental scientific advances combine with practical engineering to solve complex reservoir challenges. Each technical hurdle overcome has expanded the range of treatable reservoirs and improved recovery efficiency, contributing to more sustainable oil production.

Chemical Deep Conformance Application, Key Considerations

The following aspects are considered critical in ensuring the technical and economic success of planning chemical deep conformance applications.

Geological Heterogeneity in Carbonate Reservoirs

Carbonate reservoirs can be fairly heterogeneous due to their depositional and diagenetic history. Unlike clastic reservoirs, carbonates can exhibit appreciable variations in porosity and permeability over short distances, influenced by factors such as facies changes, dissolution features, vuggy features, and sometimes fracturing. Analysis and integration of seismic attributes, core data, log data, dynamic injection and production data are critical. The aim is to help identify and verify potential location or spatial distribution of fast conductive pathways (from water injector to producer).

The presence of compartmentalization or baffles, if any, deserves similar amount of attention. This can place a significant amount of influence on chemical deep conformance treatment feasibility and design.

Formation Porosity & Pore-Throat Architecture

Where particle-gels type of chemical deep conformance technologies are involved, it is essential to understand formation pore throat size distribution. It is a must-have input parameter in treatment design.

Fracture Intensity & Connectivity

Open flowing fractures act as high-conductivity pathways, potentially leading to gel washout if untreated

Formation Temperature and Salinity (hardness) Effects

Reservoir formation temperature measurement needs to be available. Formation temperature and salinity (or injection and formation water composition, fluid mixing) arguably have the most significant impact on chemical stability, i.e., a direct impact upon the success or not of planned treatment success.

Wettability Effects

Carbonate reservoirs often exhibit oil-wet or mixed-wet conditions, which can hinder (water-based) gel adhesion and stability. Advanced laboratory tests can be included to help ensuring this type of risks are eliminated.

Mineralogy (Clay, Anhydrite, Calcite for example)

Reactive minerals (e.g., anhydrite) can alter certain gel chemistry, significantly reducing effectiveness.

Residual Oil Saturation (ROS) and Target Identification

Chemical deep conformance treatment candidate well selection process should start with definition of selection criteria. It is recommended for remaining oil saturation distribution to be part of the selection criteria. There must be sufficient remaining oil for a well to be considered a potential treatment target. Very often, PLT logs and saturation logs can help remaining oil saturation assessment.

Reservoir Pressure, Injection and Production Rates

It is recommended to cross check that planned chemical treatment are not adversely affected by prevailing reservoir pressure, injection and production rates.

Economics Range

Based on reservoir simulation or analogue wells (ideally from the same reservoir/field), potential oil production gains must be assessed. It is recommended to consider potential uncertainties, from geological heterogeneity, reservoir properties, fluid saturation distribution, chemical performances etc., to arrive at a range of potential outcomes. This will help to provide insights into range of economical returns. If economic return is poor due to certain uncertainties, consider acquiring additional field data to reduce uncertainty range, and reassess economics outcome before a decision is made.

Chemical Deep Conformance Application, Recommended Workflow

Based on an analysis of field applications, to help improving chemical deep conformance treatment successes, a rigorous multi-disciplinary technical workflow is recommended. Key components of such a workflow are described below.

Identifying the Need for Chemical Deep Conformance

The decision to implement chemical deep conformance technology must be established by comprehensive reservoir performance analysis. Several diagnostic indicators can be deployed.

Water cut trends analysis provides the first indication of conformance issues, where systematic evaluation of water cut evolution across the field can reveal characteristic patterns. In carbonate reservoirs, diagnostic plots of water cut versus cumulative production often show hyperbolic increases characteristic of fracture-dominated flow, step-function changes suggesting sudden connection to new water sources, or early breakthrough occurring in less than a certain figure of expected flood front arrival time. Tracer tests and pressure transient analysis can be utilized for assessing potential injector to producer connections, and reservoir thickness-permeability product variations. Production logging and injection profile logging often provides the most direct evidence of water entry (at injection well) and water production profiles. Fine scale sector models should be utilized where available - to help assessing injection water pathways, and areas where remaining oil exist in sufficient quantity.

A systematic approach ensures objective identification of reservoirs areas that would benefit from chemical deep conformance implementation.

Feasibility Assessment

A rigorous feasibility assessment bridges the gap between identifying conformance problems and designing solutions. If needed, the assessment can look into potential technical topics including:

Reservoir Architecture Characterization. Consider characterization techniques below if applicable

- Advanced characterization techniques are required for carbonates:

- Micro-CT scanning of cores
- NMR T2-T1-T2 maps for pore-body to pore-throat size distributions
- Outcrop analog studies for fracture network statistics

Fluid-Rock Interaction Analysis. Geo-mechanical Assessment
Chemical Compatibility Testing

Where applicable, standard tests may be expanded to include:

- High-pressure/high-temperature rheology
- Long-term stability tests (6-12 month duration)
- Coreflood experiments with live crude at reservoir conditions
- X-ray diffraction of potential precipitates.

Economic Modeling. Risk Assessment Framework

A quantitative risk assessment matrix should be developed. The evaluation can consider, for example, geological uncertainty, chemical performance risks, operational constraints & challenges.

The feasibility assessment should produce a Go/No-Go decision supported by:

- Technical feasibility score
- NPV
- Risk-adjusted IRR
- Ability to execute and field monitoring.

Candidate Well Screening and Selection

A candidate well selection criteria must be established. In addition to reservoir production performances (i.e., sufficient remaining by-passed oil), there is a strong need to consider well mechanical integrity, for example, candidates with waterchanneling behind casing in the treatment zone must be excluded.

It is also preferred to assess operational feasibility at this stage in candidate well screening selection, for example, well accessibility, surface space constraints, HSE, workover rig availability if relevant.

Candidate Technology Selection: Screening and Selection Criteria

Candidate chemical deep conformance technology screening and selection is one of the most critical element. Although analogues can be referenced - which very often provides confidence, a thorough screening and selection exercise must be carried out. Below are a list of some of the important factors to be considered in the screening and selection:

Reservoir Condition Matching

- Temperature stability range
- Salinity tolerance
- pH operating window
- Shear resistance.

Chemical Performance Characteristics

- Gelation time control ($\pm 10\%$ target)
- Post-gel strength (elastic modulus)
- Resistance to degradation

Operational Practicalities

- Surface mixing requirements
- Pumping pressure constraints
- Onsite QA/QC capabilities
- Cleanup procedures

Economic Considerations

- Chemical and material cost per barrel treated
- Equipment requirements
- Treatment duration, manpower requirements
- Expected duration where treatment positive impact lasts (incremental oil production).

Risks

- Formation damage potential
- Environmental impact
- Vendor/supplier track record.

Detailed Laboratory Chemical Testing Program

Detailed laboratory test program must be carefully planned and executed. The test program may include: reservoir fluid characterization, chemical screening, core flood experiments. Some of the tests should be long term.

Field Pilot (First Trial)

It can be a good idea to perform a field trial on (for example) one candidate well first. The field trial will test pre-treatment preparations, treatment design, and execution. It is recommended to carry out full reservoir monitoring, to help acquiring data for assessing treatment effectiveness vs planned target. The monitoring should be designed to help reduce high impact uncertainties. Contingency plan should be drawn up, covering potential likely events.

Field Pilot/Trial Data Interpretation, Reservoir Simulation

Once field pilot trial data and results become available, fine scale sector reservoir simulation models can be developed to help with data interpretation. Based on reservoir simulation and analogues if available, input parameters can be generated for large scale expansion assessment.

Chemical formula adjustments can be made based on field trial results, with additional laboratory tests if necessary.

Large Scale Expansion Plan

Large scale expansion study can follow a similar approach to the field pilot/trial design, with field trial input and any adjustment to the chemicals.

Technical sophistication relating to the method used to size the volumes of various conformance-improvement gel treatments can be further improved. Although sizing of gel treatments remains highly empirical in new field applications, there are a number of empirical guidelines for sizing conformance gel treatments.

A fit for purpose reservoir monitoring plan should be generated and implemented, to ensure treatment effectiveness. Cost optimization opportunities should be identified and pursued where applicable.

Field Case Examples

Figure 1 shows chemical deep conformance water shut off application in an extremely challenging fractured vuggy carbonate reservoirs. For the fractures, a delayed in-situ polymerized gel with enhanced stability compared to conventional gels was developed, applicable for temperatures up to 140°C, salinity up to 220,000 mg/L, and calcium-magnesium ions up to 10,000 mg/L. Gelation time was significantly delayed from 1 hour to a controllable range of upto to 70 hours. The gel exhibits good selectivity, being oil-soluble and water-insoluble. For vugs in the reservoir: density-selective solidifying particles with a maximum temperature resistance of 190°C. The particle density is between formation water and crude oil, allowing automatic spreading at the oil-water interface in vugs to form a selective barrier.

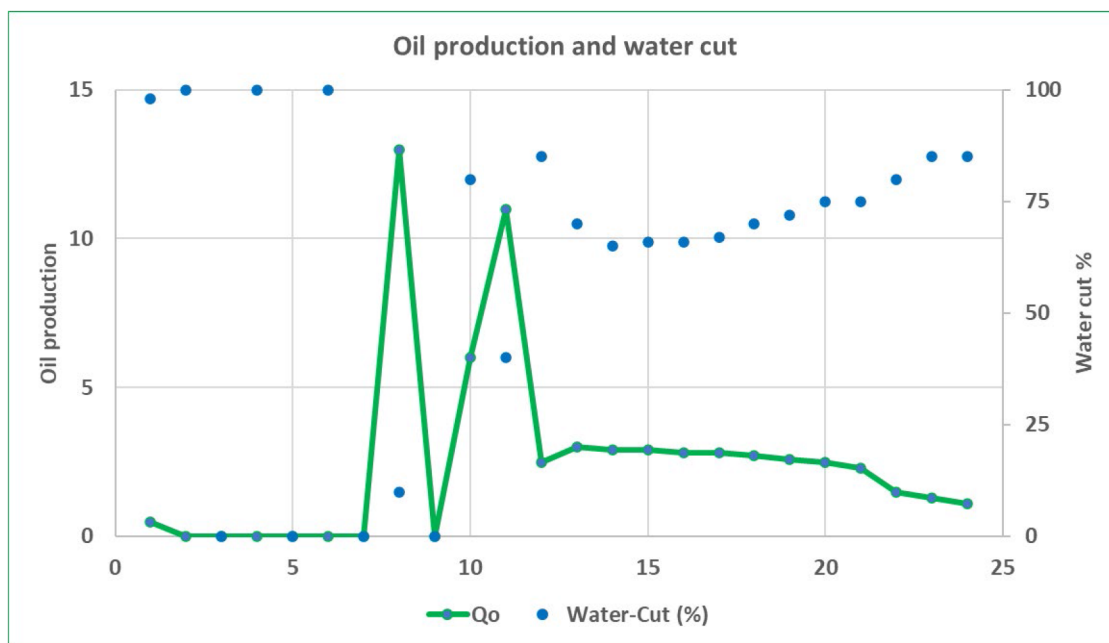


Figure 1—Well watered out and back to production after deep conformance treatment

Some 150 water shut off operations were carried out in T* field in China, all fracture vuggy carbonates, significant amount of incremental oil production was obtained.

Figure 1 shows an example from the water shutoff operations. Well TX* was put on production by depletion, water cut increased due to bottom water coning. The well was shut in to reduce coning effect: which was successful in the first time, but becoming less effective subsequently. The well was put on ESP, but production showed rapid decline due to water cut increase. The well was shut in being considered fully watered out.

A chemical deep conformance treatment plan was designed and carried out, bearing in mind the presence of fractures, vugs and aquifer water drive. Chemicals were pumped at 20 to 50 bbls/minute, maximum pumping pressure reached 1500 psi at surface. Designed volume was pumped successfully.

After treatment, the well was put back on production. Oil production rate came in at expected range, while water cut dropped to 50%, indicating a successful chemical deep conformance treatment. The treatment effect lasted more than 4 months, with oil production revenue more than offsetting total costs.

Field case 2 is a low permeability faulted reservoir with well developed fractures. The reservoir was developed with water flooding. Some 40% of the production wells showed accelerated water cut development, with well head formation water sampling analysis indicating injection water breakthrough. Chemical conformance treatments were initiated in the early 2010's, but the success rate was relatively low, and incremental oil insufficient to justify the incurred operational costs.

Further research and development were conducted to improve the conformance control agents formula targeting waterchanneling fractures. It was found that a single treatment was insufficient, a multi-round conformance treatment approach was adopted to help ensuring longer lasting effects. Post treatment well head production water sampling and analysis showed a decline in chloride indicating treatment success (due to the relatively close well spacing). Production well water cut reduction by 25% was observed. Production water cut continue to be below pre-treatment level for up to one year.

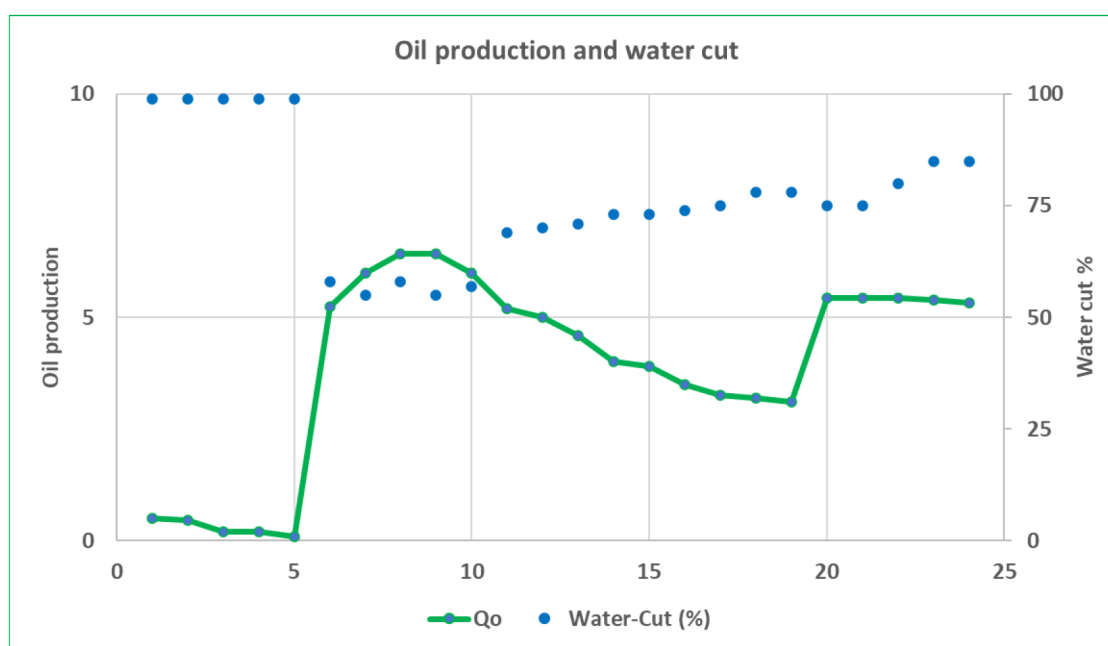


Figure 2—Fractured low permeability reservoir, multi treatment field case

Field case 3 is a moderate permeability reservoir with identified micro-fracture networks. Chemical deep conformance was successful in reducing water cut by up to 50%. See [Figure 3](#).

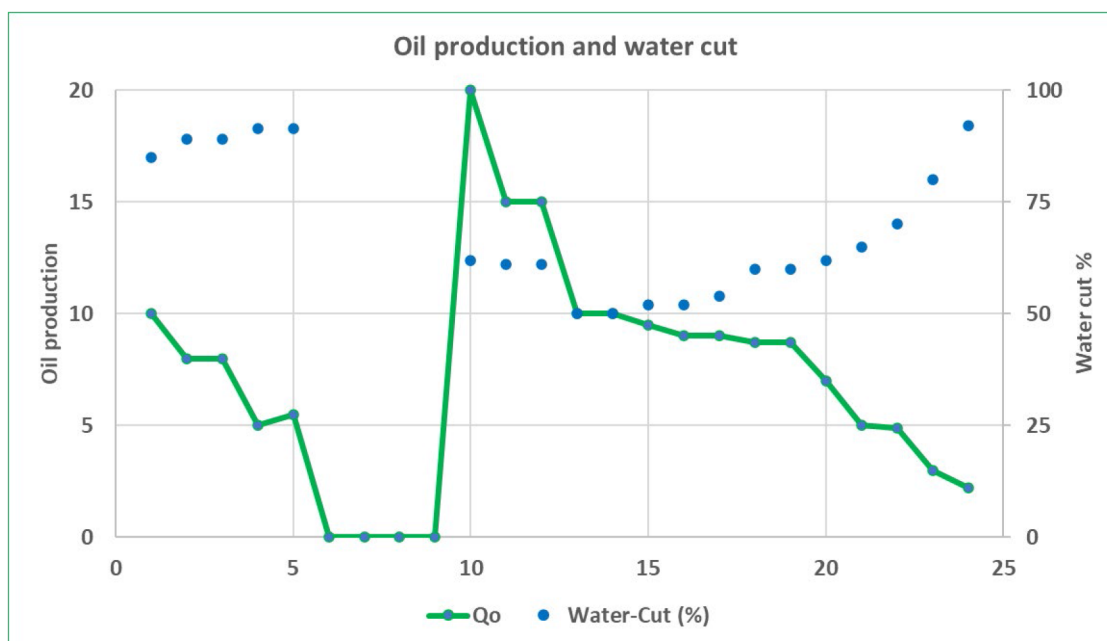


Figure 3—Carbonate reservoir, possible direct fracture connection from injector to producer

Field case 4 is from a reservoir with bottom aquifer water drive. The horizontal production well was completed with sand screen. It was difficult to identify water breakthrough points, treatment was bull headed assuming the most watered out section would take in the bulk of the treatment. Figure 4 shows that the treatment has been successful with water cut reduction of 35%.

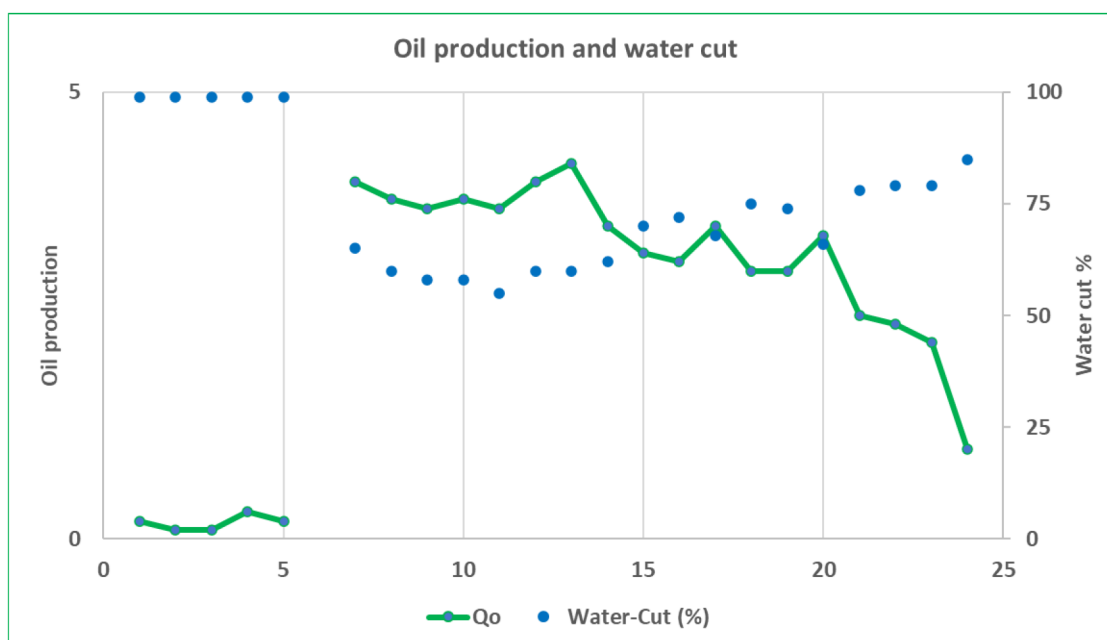


Figure 4—Carbonate reservoir, horizontal well with bottom water breakthrough

Injectivity Management

Chemical deep conformance can be applied in both production well and injection wells. In carbonate reservoir development, horizontal water injection wells can have higher injectivity than a vertical injection well, and horizontal injection wells can be utilized to target certain reservoir section/zone. Maintaining

and managing injection well injectivity, injection profile, should go hand in hand with any chemical deep conformance in (flow connected) production wells in order to maximise oil recovery and arrest potential oil production decline.

One of the key challenges in water injection through horizontal wells is achieving uniform distribution along the wellbore. Non-uniform injection can lead to premature water breakthrough, poor sweep efficiency, and reduced recovery. Therefore, ensuring uniform water injection distribution is critical for optimizing reservoir performance, extending field life, and maximizing economic returns.

Uniform water injection can help to ensure that the injected water displaces oil evenly across the reservoir, preventing preferential flow paths that can lead to early water breakthrough in certain sections of the well. Carbonate reservoirs often exhibit high permeability streaks, fractures, or vuggy porosity, which can cause injected water to channel through these high-permeability zones, bypassing oil-saturated regions. When water injection is uniform, the sweep efficiency improves, allowing more oil to be mobilized and produced before water cuts rise excessively. This can lead to higher ultimate recovery factors, which is a primary objective in reservoir management.

Non-uniform water injection can result in localized water breakthrough, where water rapidly reaches the production wells through high-permeability zones. This leads to increased water production, requiring additional processing and disposal costs. Moreover, excessive water cycling (where injected water quickly circulates between injectors and producers without effectively displacing oil) reduces the efficiency of the waterflooding project. By maintaining uniform injection, the injected water advances more uniformly, delaying breakthrough and ensuring that a larger portion of the reservoir is effectively swept before water reaches the producers.

Water injection serves not only to displace oil but also to maintain reservoir pressure, likely preventing rapid declines that can reduce well productivity. Uniform water injection should not sacrifice good injectivity section of the well - leading to potentially lower voidage replacement ratio, and insufficient pressure support.

One of the emerging technology for help creating uniform water injection while not jeopardizing overall horizontal well injectivity is Foam Acid treatment. Foam acid is a dispersion system in which gas is stabilized in acid liquid by a foaming agent. The gas phase consists of nitrogen supplied by nitrogen generation equipment, while the liquid phase is composed of various acid solutions tailored to the well conditions. The foamed acid is pumped into high-permeability water-bearing zones, gradually increasing fluid flow resistance and creating a gas-blocking effect in the pore throats.

Under the combined effect of this gas blockage, the foamed acid then enters low-permeability formations to react with the rock, generating additional etching channels. This helps remove contamination and blockages in low-permeability layers, improving the fluid production profile of the well. Finally, a foam-assisted acid flowback fluid is injected to assist in flowback and induce well kick, ensuring the removal of spent acid.

The T* Oilfield, located in the northwestern region of China's Tarim Basin, represents one of the most technically challenging carbonate reservoirs in the country. Characterized by extreme reservoir conditions including deep burial depths ranging from 5,000 to 7,000 meters, high bottomhole temperatures between 120-150°C, and significant heterogeneity with permeability variations from less than 1 mD in the matrix to 50 mD in fracture networks, this field has presented persistent stimulation challenges. Conventional acidizing treatments in such environments often result in inefficient fluid distribution, with the majority of acid preferentially entering high-permeability zones while leaving the tighter matrix sections untreated. This case study examines the successful application of foam acid stimulation technology in Well TXX, a horizontal producer that had demonstrated suboptimal performance following previous stimulation attempts.

The subject well, TXX, was completed in the Ordovician carbonate formation at a measured depth of 6,200 meters with a 800-meter horizontal section. Prior to the foam acid treatment, the well was producing

approximately 12 m³/day of oil with a 65% water cut, significantly below the expected production rates for this reservoir segment. Diagnostic production logging runs revealed that nearly 80% of the inflow was coming from just 20% of the completed interval, indicating severe uneven production contribution along the lateral. Pressure transient analysis suggested the presence of both natural fractures and matrix porosity, with permeability contrasts exceeding two orders of magnitude between different sections of the wellbore.

The foam acid treatment was specifically designed to address these challenges through improved fluid diversion and more uniform stimulation across the heterogeneous formation. The fluid system consisted of a 20% hydrochloric acid base enhanced with specialized additives including high-temperature corrosion inhibitors, iron control agents, and a thermally stable foaming surfactant specifically formulated for the harsh downhole conditions. The gas phase utilized nitrogen generated onsite through a membrane separation unit, providing consistent gas supply at the required rates and pressures. The foam quality was maintained between 70-75% throughout the treatment, a range determined through laboratory testing to provide optimal stability and diversion capability under reservoir conditions.

The treatment procedure was carefully sequenced to maximize effectiveness. An initial pre-flush stage consisting of a foam diverter slug was pumped to temporarily block the high-permeability zones and fracture networks. This was followed by the main foam acid stage, where the foamed acid was injected at a controlled rate of 2.5 m³/min while maintaining constant nitrogen co-injection. The foam's unique rheological properties created selective diversion within the formation, effectively forcing the acid into the lower permeability matrix sections that had been bypassed in previous treatments. Real-time monitoring of treating pressures and fluid distribution was achieved through distributed temperature sensing (DTS) and downhole pressure gauges, allowing for immediate adjustments to the treatment parameters as needed.

Following the acid reaction period, a specially formulated foam-assisted flowback fluid was pumped to facilitate efficient cleanup of the well. This fluid system combined nitrogen foam with a viscosifier to enhance lifting capacity while minimizing formation damage during the flowback process. The well was then brought back on production using a controlled, closed flowback system to properly clean up the entire stimulated interval and evaluate treatment effectiveness.

The results of the foam acid treatment demonstrated significant improvements across all key performance indicators. Oil production increased from the pre-stimulation rate of 3 fold, representing a fourfold improvement over the baseline and nearly doubling the best results achieved with previous conventional acidizing treatments. Perhaps equally importantly, the water cut decreased from 65% to 45%, indicating that the treatment had successfully addressed the water channeling issues that had plagued the well's performance. Bottomhole flowing pressure increased from 28 MPa to 38 MPa, reflecting improved reservoir connectivity and energy support. Production logging conducted three months after the treatment showed a much more uniform production profile along the lateral, with over 70% of the completed interval now contributing to flow compared to just 20% prior to treatment.

The success of this treatment can be attributed to several key factors. The foam system's ability to temporarily block high-permeability pathways forced the acid into previously under-stimulated zones, creating a more uniform distribution of conductive channels throughout the formation. The nitrogen component provided enhanced energy for flowback, ensuring effective cleanup of spent acid and reaction products from the entire treated interval. The high-temperature stability of the foaming system maintained its integrity throughout the treatment, even under the extreme downhole conditions. Perhaps most significantly, the production increase proved to be sustainable, with the well maintaining stabilized production rates for over 12 months post-treatment, compared to the typical 4-6 month sustainability seen with conventional acidizing in this field.

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